

Latent Planning for Long-Horizon Event Prediction

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Visual overview

Long-horizon prediction requires a model to preserve useful information and decide how far into the future to reason. The Predictron learns a latent Markov reward process and blends predictions from several internally imagined planning depths.

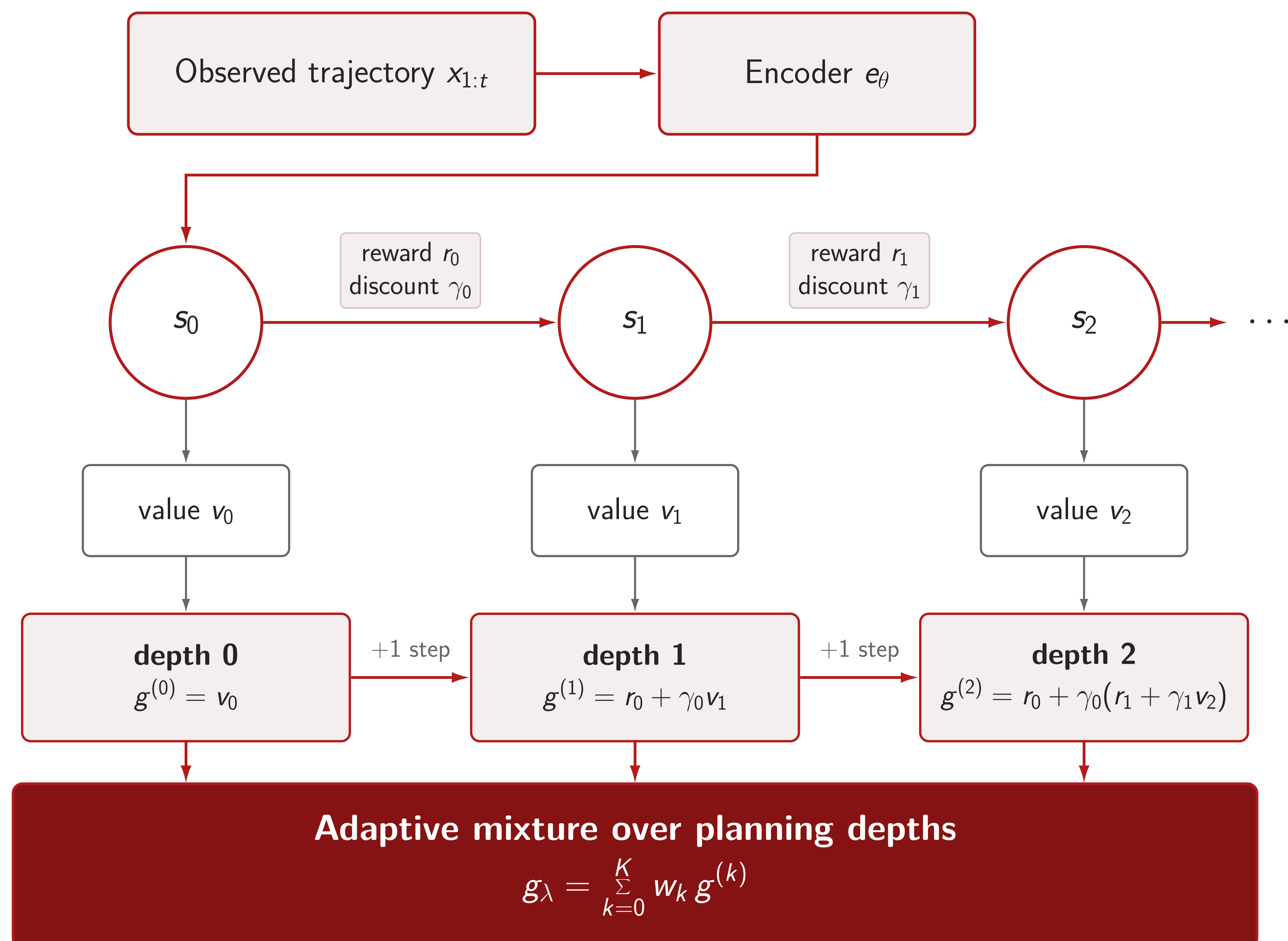


Figure 1: Latent planning architecture. Adapted from Silver, D., Hasselt, H., Hessel, M., Schaul, T., Guez, A., Harley, T., & Degris, T. (2017, July). The predictron: End-to-end learning and planning. In *International Conference on Machine Learning* (pp. 3191–3199). PMLR.

Predictron

Describe data, model, or experimental setup.

Methods

Describe data, model, or experimental setup.

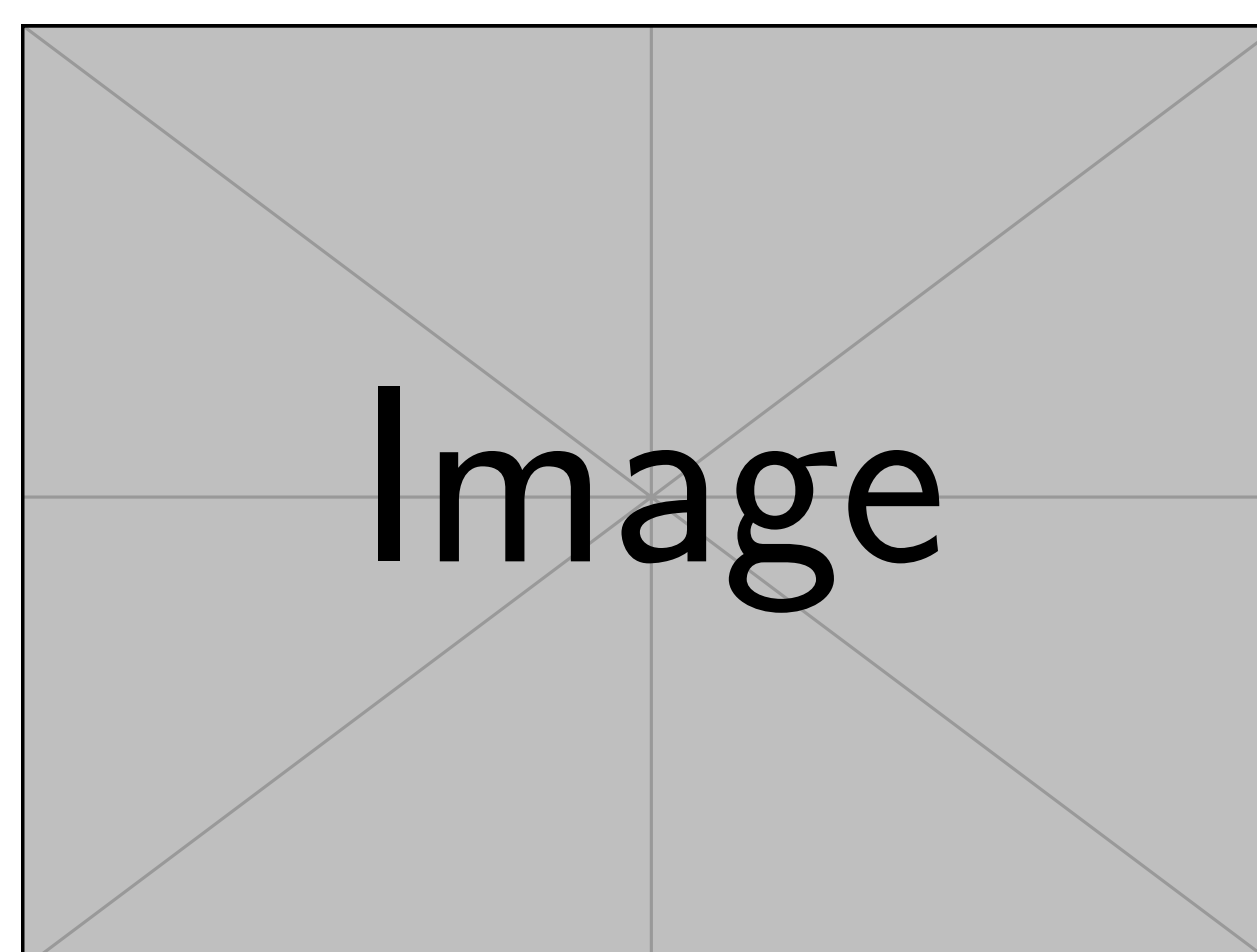
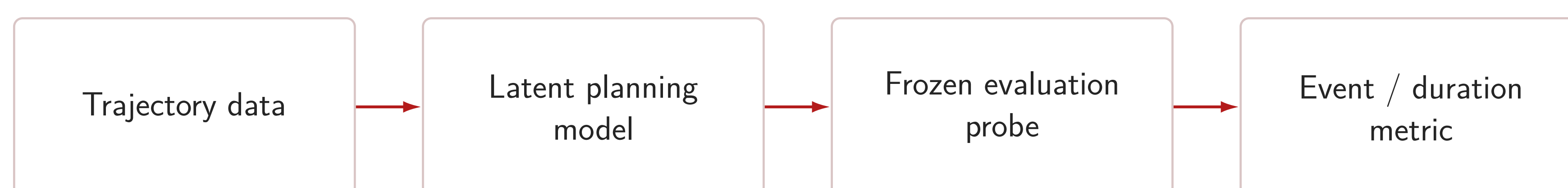


Figure 2: Overview of the proposed pipeline.

Experimental pipeline



Introduction

Motivate the problem in 3–5 sentences. State why it matters and what gap your work fills.

- Key point one
- Key point two

Markov reward process (MRP)

An MRP is defined by:

$$s', r, \gamma = p(s, \alpha) \quad (1)$$

- s' : Next state
- r : Reward
- γ : Discount factor

Equations

$$\hat{y} = \arg \max p(y | x; \theta) \quad (2)$$

Results

Method	Accuracy	F1
Baseline	0.812	0.795
Ours	0.904	0.891

Table 1: Comparison on the test set.

Conclusions

- Main takeaway
- Limitation and future work

References

[1] Author, A. (2024). Title of the paper. *Journal*, 12(3), 45–67.